

What's on My Skin?



I. Project Name and Description

"What's on My Skin?" is an **AI algorithm-based** facial skin disease prediction and treatment suggestion system.

I've developed a system that allows patients with skin conditions to upload a facial photo and determine whether their skin is normal, has acne, acne scars, or hyperpigmentation.

After analyzing the patient's skin condition, the system provides the analysis results along with solutions for each specific skin disease.

II. Project Duration and Milestones

- 24.07.15 ~ 24.07.28: Investigated detailed classification of each symptoms and identified solutions; gathered facial skin disease datasets for each symptom.
- 24.07.29 ~ 24.07.31: Labeled datasets, performed data preprocessing and augmentation.
- 24.08.02 ~ 24.08.15: Conducted model development and training, followed by testing and performance evaluation to optimize the model.
- 24.08.16 ~ 24.08.22: Developed a demo version of the webservice.

III. Technology Stacks Used

LANGUAGE



AI MODEL

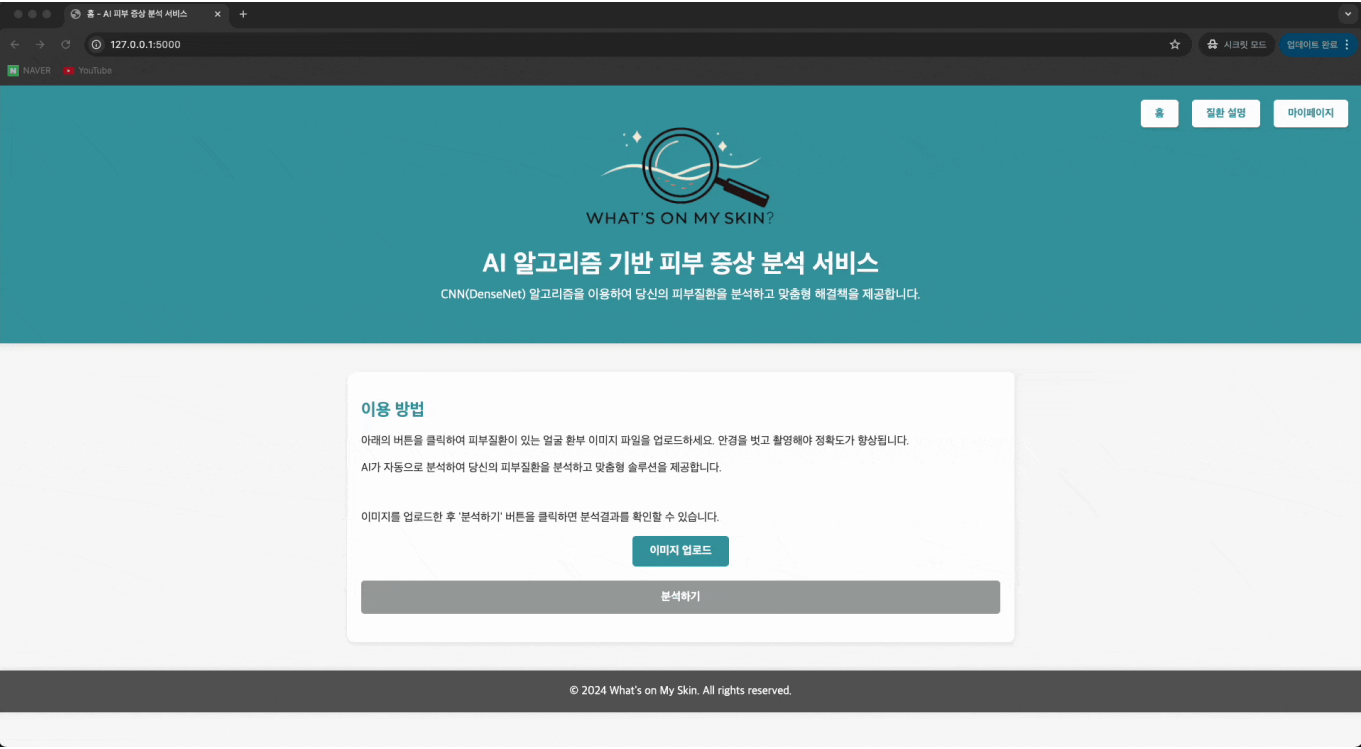


WEBSERVICE



IV. System Usage

1. Localhost service (for demonstration)



2. Beta webservice using streamlit

 [WEBSITE LINK : whatisonmyskin.streamlit.app](https://whatisonmyskin.streamlit.app)

- Please use the website in light mode.



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AI 알고리즘 기반 피부 증상 분석 서비스

⚠️ 주의사항: 분석 결과는 참고용이며, 정확한 진단 및 치료를 위해 반드시 피부과 전문의와 상담하시기 바랍니다.

📁 이미지 파일을 업로드하세요 📁



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V. Installation (for localhost test)

1. Clone the repository :

```
git clone https://github.com/IJLee0812/What-is-on-My-Skin.git
```

2. Start the web server :

```
cd What-is-on-My-Skin/webservice
```

```
pip install Flask Pillow torch torchvision -> Commands to install the  
tools and libraries required to run app.py
```

```
python3 app.py
```

3. Open the webservice in your browser :

```
http://127.0.0.1:5000 or http://localhost:5000
```

VI. Key Development Explanation (AI)

1. About Data

- I **manually collected** 250 facial images for each of the three skin diseases as well as normal skin. Below are the four class names and Korean translations that the AI model must classify.
 - acne(여드름)
 - acne_scar(여드름 흉터)
 - hyperpigmentation(색소침착)
 - normal(정상 피부)
- Due to the limitations in the dataset size that an individual can gather, data augmentation and 5-Fold Validation techniques were applied.

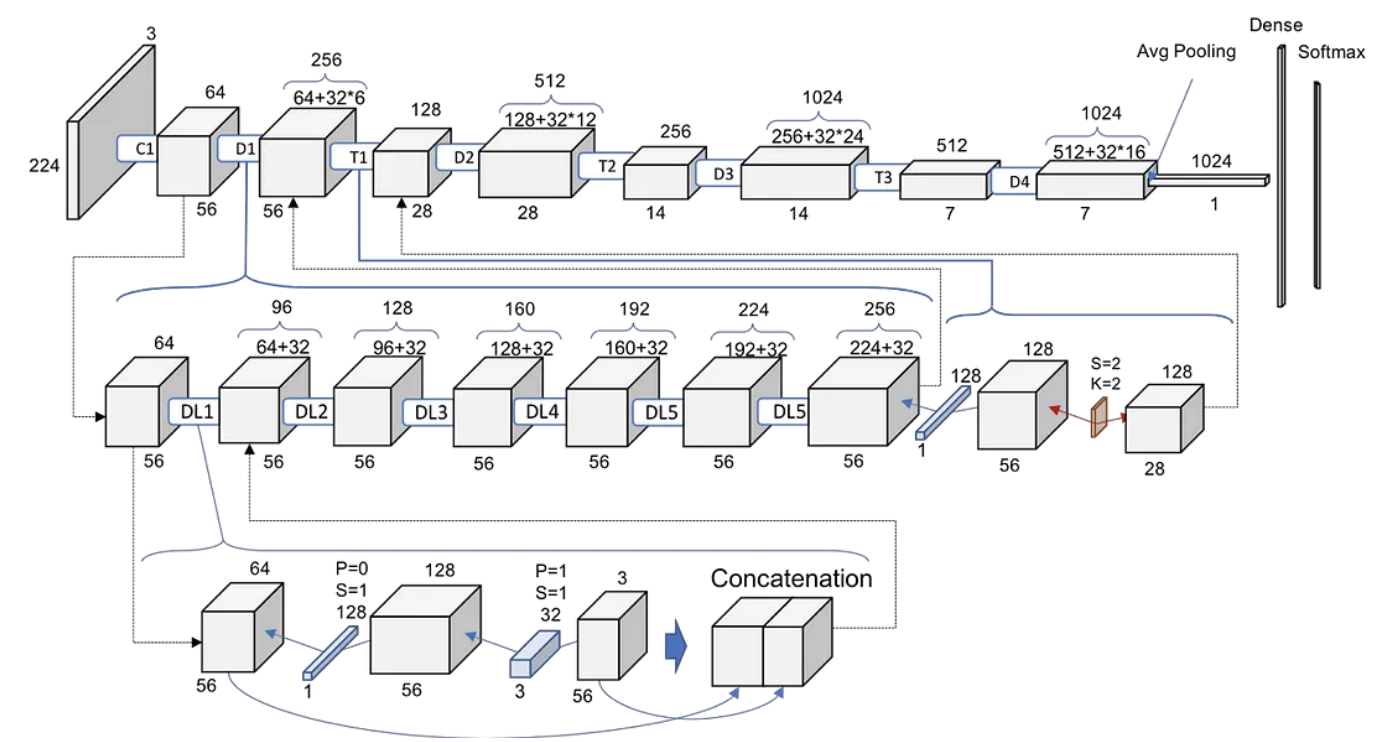
2. Data Preprocessing & Augmentation

```
CenterCrop(224) -> RandomHorizontalFlip() -> RandomRotation(degrees = 20)  
-> ToTensor() -> Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
```

- CenterCrop(224): This operation resizes the input image by cropping it to 224x224 pixels from the center. This ensures that the key features of the image are preserved.
- RandomHorizontalFlip(): The image is randomly flipped horizontally with a 50% probability. This technique helps the model generalize better by introducing variations in the orientation of the images.
- RandomRotation(degrees = 20): The image is rotated randomly within a range of ± 20 degrees. This augmentation technique is useful for making the model invariant to slight rotations in the input images.
- ToTensor(): This transforms the image into a PyTorch tensor, where the pixel values are scaled to a range between 0 and 1. Tensors are the primary data structure used in PyTorch models.
- Normalize([0.485, 0.456, 0.406], [0.229, 0.224, 0.225]): The image tensor is normalized using a mean of [0.485, 0.456, 0.406] and a standard deviation of [0.229, 0.224, 0.225]. These values are standard for models pretrained on ImageNet, helping to standardize the input data.

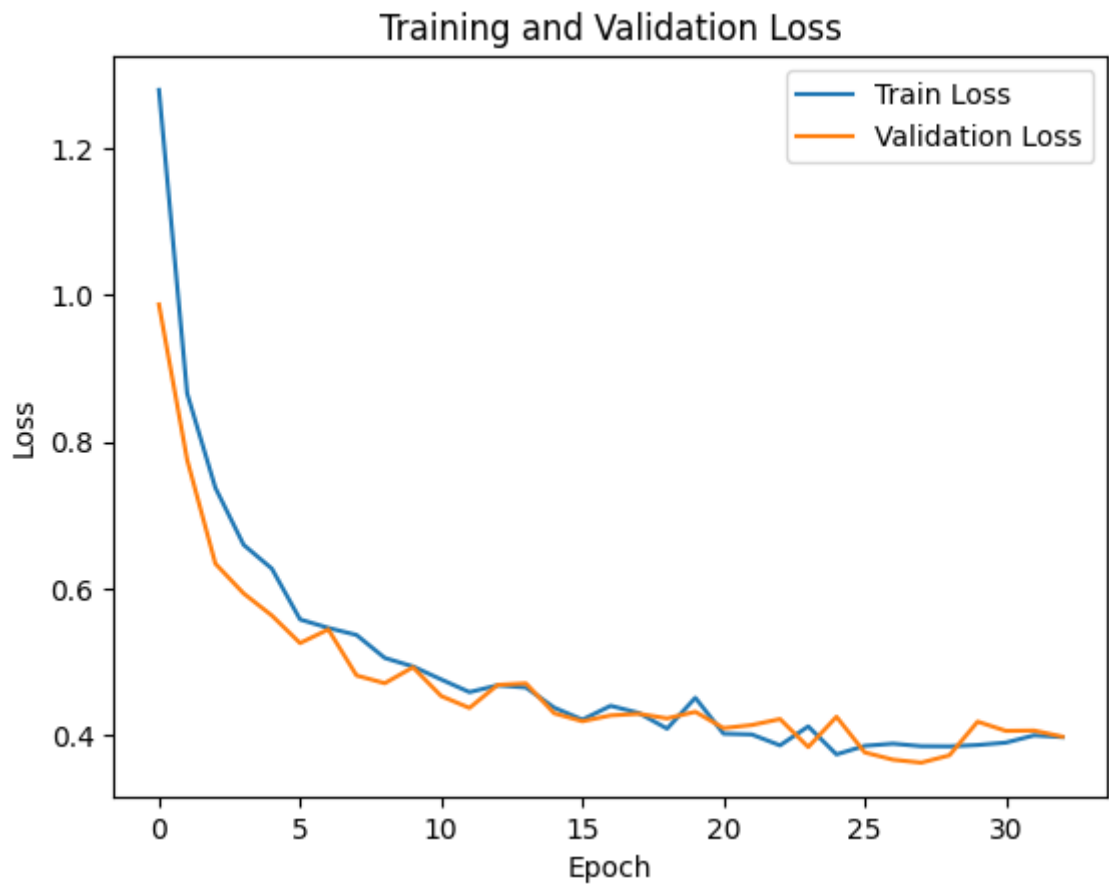
3. AI Training

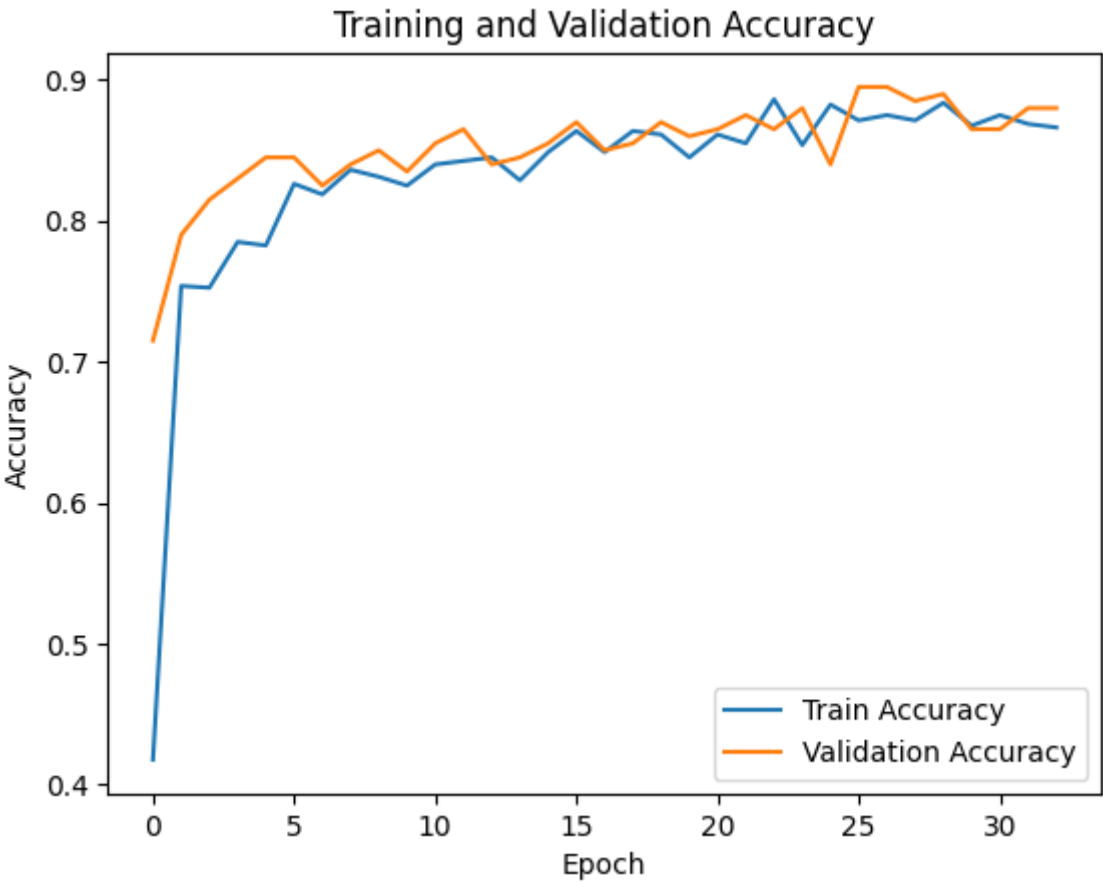
- The **CNN** algorithm used was a **pretrained DenseNet121**. Below is a schematic diagram of the structure of DenseNet.



- To advance training, Early Stopping and LR scheduling techniques were applied.
 - LR Scheduler : ReduceLROnPlateau

4. Model Performance Evaluation





The model achieved meaningful level of accuracy in both training and validation (around 88%), indicating strong performance with minimal overfitting.

The training process effectively increased accuracy, suggesting that learning has largely stabilized, and further epochs may not yield significant improvements.

